Exam Equations

Trigonometry: $\sin\Theta = h_o/h$ $\Theta = \sin^{-1}(h_o/h)$ $\cos\Theta = h_a/h$ $\Theta = \cos^{-1}(h_a/h)$ $\tan\Theta = h_o/h_a$ $\Theta = \tan^{-1}(h_o/h_a)$

Pythagorean Theorem $h^2 = h_o^2 + h_a^2$

$\cos(90° - \Theta) = \sin(\Theta)$ $\cos(90° + \Theta) = -\sin(\Theta)$ $\cos(180° - \Theta) = -\cos(\Theta)$
$\sin(90° - \Theta) = \cos(\Theta)$ $\sin(90° + \Theta) = \cos(\Theta)$ $\sin(180° - \Theta) = \sin(\Theta)$

Vectors: $\vec{\mathbf{a}} = (a_x)\hat{\mathbf{x}} + (a_y)\hat{\mathbf{y}}$ $\vec{\mathbf{a}} \bullet \hat{\boldsymbol{x}} = a_x = |\vec{a}|\cos(\alpha)$ $\vec{\mathbf{a}} \bullet \hat{\boldsymbol{y}} = a_y = |\vec{a}|\cos(\beta)$

$$\alpha = \cos^{-1}\left[\frac{a_x}{|\vec{\mathbf{a}}|}\right] \qquad \beta = \cos^{-1}\left[\frac{a_y}{|\vec{\mathbf{a}}|}\right] \qquad |\vec{\mathbf{a}}| = \sqrt{\vec{\mathbf{a}} \bullet \vec{\mathbf{a}}} = \sqrt{(a_x)^2 + (a_y)^2}$$

Dot product: $\vec{a} \bullet \vec{b} = |\vec{\mathbf{a}}||\vec{\mathbf{b}}|\cos(\varphi)$ Cross Product: $\vec{c} = \vec{a}x\vec{b} = \left[|\vec{\mathbf{a}}||\vec{\mathbf{b}}|\sin(\varphi)\right]\hat{c}$

Displacement: $\vec{\boldsymbol{x}} = \vec{r}_f - \vec{r}_i$ Speed = d/t velocity: $\vec{\mathbf{v}} = d\vec{\mathbf{x}}/dt$ acceleration: $\vec{a} = d\vec{\mathbf{v}}/dt$

Newton's Laws of Motion $\Sigma\vec{\mathbf{F}} = m\vec{\boldsymbol{a}}$ or $\vec{\boldsymbol{a}} = \Sigma\vec{\boldsymbol{F}}/m$ Weight: $\vec{W} = m\vec{\boldsymbol{g}}$, $|\vec{\boldsymbol{g}}| = 9.8$ N/kg

$\Sigma\vec{\mathbf{F}} = \vec{\mathbf{F}}_{\mathbf{net}} = (\Sigma F_x)\hat{\boldsymbol{x}} + (\Sigma F_y)\hat{\boldsymbol{y}}$ where $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$

For equilibrium: $\Sigma F_x = 0$ and $\Sigma F_y = 0$ For non-equilibrium: $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$

Centripetal Acceleration and Force, $a_c = v^2/r = r\omega^2$, $F_c = ma_c = m\,v^2/r$

Kinetic Energy: $KE = \frac{1}{2}mv^2$ Gravitational Potential Energy: $PE_g = mg(y_f - y_i) = mgh$

Elastic Potential Energy: $PE_e = \frac{1}{2}kx^2$, Springs: $F = k\Delta x$,, Conservation of Energy, $\Delta KE + \Delta PE = 0$

Work /Energy, $W = \Delta KE + \Delta PE = (F\cos\varphi)\Delta x$ Total work: $W_{net} = W_1 + W_2 + W_3 + \ldots$ or $W_{net} = (\Sigma F)x$

Potential Difference for uniform electric field: $\Delta V = -E\Delta x$ Work in uniform electric field: $W = qE\Delta x$

Work, Potential and Potential Energy: $\Delta V = PE_E/q$, $W = -\Delta PE_E$ $\Delta PE_E = q\Delta V$ Point charge: $PE_E = kq_1q_2/r$

Electric Potential for point charge: $V = kq/r$ relative to ∞, Potential Difference for point charge: $\Delta V = kq/r_1 - kq/r_2$,

Total Potential for point charge: $V_T = V_1 + V_2 + V_3 + \ldots$

$$Work:\ W = \int_A^B \vec{F} \bullet d\vec{x} = -\Delta PE_E = \int_A^B \left[k\frac{Qq}{r^2}\right]\hat{r} \bullet d\vec{x} = -kqQ\left[\frac{1}{r_B} - \frac{1}{r_A}\right] \quad W = -\Delta PE_E$$

$$W = PE_{system} = PE_{1st} + PE_{2nd} + PE_{3rd} + \cdots$$

$$\vec{E} = -\nabla V = -\left[\left(\frac{\partial V}{\partial x}\right)\hat{x} + \left(\frac{\partial V}{\partial y}\right)\hat{y} + \left(\frac{\partial V}{\partial z}\right)\hat{z}\right] \qquad \Delta V = V_B - V_A = -\int_A^B \vec{E} \bullet d\vec{x} \qquad V = k\int \frac{1}{r}dq$$

Linear charge density: $\lambda = Q/L$ $dq = \lambda dL$ Surface charge density: $\sigma = Q/A$ $dq = \sigma dA$

Volume charge density: $\rho = Q/V$ $dq = \rho dV$

Sphere: $V_{Sphere} = \frac{4}{3}\pi R^3$ $A_{Sphere} = 4\pi R^2$ Cylinder: $V_{Cylinder} = \pi LR^2$ $A_{Cylinder} = 2\pi RL + 2\pi R^2$
Rectangular prism: $V_{Rectanglar} = Lwh$ $A_{Rectagular} = 2(Lw + Lh + wh)$
Area of circle: $A_{circle} = \pi r^2$ Circumference of Circle: $C_{circle} = 2\pi r$ Area of a right triangle: $A_{Triangle} = \frac{1}{2}hb$
Angle Θ for circle: $\Theta = s/R$

Charge on a capacitor $q = CV$, $q = (\kappa\varepsilon_o A/d)\Delta V$, Capacitor and electric field: $\Delta V = E\Delta x$

Dielectric Constant: $\kappa = E_o/E = C/C_o = V_o/V$, $\kappa = 1.0$ for vacuum or air

Capacitance of Capacitor $C = \kappa\varepsilon_o A/d$, $C = \kappa C_o$, $E_o = q/(\varepsilon_o A)$, Stored Energy of Cap: Energy $= \frac{1}{2}C(\Delta V)^2$

Current, $I = dq/dt$, Charge: $Q = Ne$, Ohm's Law, $V = IR$, Power, $P = IV = I^2R = V^2/R$, P = Energy/time

Peak and rms current/voltage: $I_{rms} = I_p/\sqrt{2}$ $V_{rms} = V_p/\sqrt{2}$

Resistivity and Resistance: $R = \rho L/A$ $R = R_o(1+\alpha(T-T_o))$ Internal resistance of battery: $V_{ext} = V_{emf} - Ir$

$k = 9.0 \times 10^9\ Nm^2/C^2$, $\varepsilon_o = 8.85 \times 10^{-12}\ C^2/Nm^2$, $\mu_o = 1.257 \times 10^{-6}\ Tm/A$ $1\ eV = 1.6 \times 10^{-19}\ J$

Electron: $q_- = 1.6 \times 10^{-19}$ C, negative charge, $m_e = 9.1 \times 10^{-31}$ kg

Proton: $q_+ = 1.6 \times 10^{-19}$ C, positive charge, $m_p = 1.67 \times 10^{-27}$ kg

1m = 100 cm, 1 m = 1,000 mm, 1 cm = 10 mm, 1 micron (μ) = 10^{-6}, 1 nano (n) = 10^{-9}, 1 pico (p) = 10^{-12}

Rulers for Derivatives

$$Power\ rule: \ \frac{d}{dx}[x^n] = nx^{n-1}$$

$$Addition\ or\ subtraction: \ \frac{d}{dx}[y \pm u] = \frac{d}{dx}[y] \pm \frac{d}{dx}[u] \qquad Product\ rule: \ \frac{d}{dx}[yu] = u\frac{d}{dx}[y] + y\frac{d}{dx}[u]$$

$$\frac{d}{dt}[\sin(\omega t)] = \omega\cos(\omega t) \qquad \frac{d}{dt}[\cos(\omega t)] = -\omega\sin(\omega t) \qquad \frac{d}{dt}[e^{at}] = ae^{at}$$

Integration Rules

$$\int (0)dx = C \qquad \int (1)dx = x + C \qquad \int (mx)dx = m\int (x)dx \qquad where\ C, a\ and\ m\ are\ constants$$

$$Addition\ and\ subtraction: \ \int (u+v)dx = \int (u)dx + \int (v)dx$$

$$Power\ Rule \quad \int x^n\,dx = \frac{x^{n+1}}{n+1} \qquad \int \sin(mx)\,dx = -\frac{\cos(mx)}{m} \qquad \int \cos(mx)\,dx = \frac{\sin(mx)}{m} \qquad \int e^{at}dt = \frac{1}{a}e^{at}$$

$$Definate\ Integral: \ \int_a^b [Integrand]dx = [Integral]\,|_a^b = \text{Integral with (b)} - \text{Intergral with (a)}$$

$$\int \frac{1}{x}dx = ln[x] \qquad \int \frac{1}{(x^2 \pm a^2)^{\frac{1}{2}}}dx = ln\left[x + \sqrt{x^2 \pm a^2}\right] \qquad \int \frac{x}{(x^2 \pm a^2)^{\frac{1}{2}}}dx = \sqrt{x^2 \pm a^2}$$

$$\int \frac{1}{(x^2+a^2)^{3/2}}dx = \frac{x}{a^2(x^2+a^2)^{1/2}} \qquad \int \frac{x}{(x^2+a^2)^{3/2}}dx = -\frac{1}{(x^2+a^2)^{1/2}}$$